PULLURU BHAVITH

AI Engineer | LLM Applications | Multi-Agent Systems | RAG Pipelines

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience designing and deploying production-grade LLM applications, multi-agent systems, and scalable RAG pipelines. Specializes in end-to-end pipeline architecture — from intelligent OCR extraction and agentic content validation to real-time conversational AI, hybrid vector search, and human-in-the-loop automation. Proficient in LangChain, LangGraph, FastAPI, Qdrant, and cloud platforms (AWS, Azure). Delivers measurable, intelligent solutions at scale with a strong emphasis on observability, async design, and system reliability.

TECHNICAL SKILLS

Languages & Frameworks: Python, Java, SQL, FastAPI, REST APIs, Node.js, React

AI / ML / LLMs: LLMs, RAG, Hybrid RAG, Multi-Agent Systems, Prompt Engineering, Mistral OCR, GPT-4o, GPT-4o-mini, OpenAI TTS, Voyage AI Embeddings, LangChain, LangGraph, LlamaIndex, Scikit-learn, TensorFlow, Keras, Computer Vision (OpenCV)

Agentic & Protocols: LangGraph StateGraph, Human-in-the-Loop (HITL), MCP (Model Context Protocol), Tool Calling, Agent Orchestration, Async Pipeline Design

Databases & Cloud: Qdrant (Vector DB), PostgreSQL, MongoDB, MySQL, Azure Cosmos DB, AWS S3, Google Slides API, Scalekit SDK

Concepts & Tools: System Architecture, Pipeline Orchestration, Async Processing, Observability (Langfuse), OOP, Design Patterns, Docker, ReportLab, python-pptx, SearXNG

PROFESSIONAL EXPERIENCE

AI Engineer | Prowess Software Services Private Limited — Hyderabad, India *Sep 2024 – Present*

- Architected and deployed production LLM applications using LangChain, LangGraph, and OpenAI GPT-4o / GPT-4o-mini with multi-agent orchestration patterns.
- Built and maintained 40+ FastAPI REST endpoints powering AI Tutor, Debate, Seminar, Quiz, and English Teacher engines for an edtech platform.
- Designed a Hybrid RAG pipeline combining Qdrant vector search (Voyage AI embeddings) with SearXNG web search for real-time, context-enriched retrieval.
- Implemented per-student Google OAuth via Scalekit SDK and Google Slides API (batchUpdate) enabling autonomous AI-driven slide generation and correction.
- Integrated end-to-end observability with Langfuse for token usage, latency, and quality monitoring across all AI pipelines.
- Deployed cloud infrastructure on AWS S3 (TTS audio storage) and Qdrant Cloud (vector search), ensuring scalable and reliable service delivery.
- Implemented adaptive learning features: topic-aware homework auto-assignment, AI-based quiz grading, and unified student performance timeline tracking.

AI Software Engineer Intern | Tardiverse — Chennai, India *Aug 2024 – Jan 2025*

- Designed and deployed the SalesPal AI chatbot using Python, LangChain, LangGraph, and Azure OpenAI, improving customer engagement by 30%.
- Built RAG pipelines and custom AI agents for domain-specific knowledge retrieval tasks using LangChain and LlamaIndex.
- Built scalable backend systems using Azure Cosmos DB, supporting 1,000+ concurrent users with 80% uptime.
- Developed dynamic user interfaces using HTML, JavaScript, Node.js, and React, boosting application responsiveness and user engagement by 10%.

KEY PROJECTS

GradeUp AI Platform *[Production · AI/EdTech]*

End-to-end AI-powered edtech platform featuring a 14-agent LangGraph pipeline for intelligent textbook processing, autonomous slide correction, hybrid RAG retrieval, and adaptive student learning at scale.

- Architected a LangGraph multi-agent pipeline with 14 specialized agents — OCR extraction, structure discovery, blueprint builder/judge, content gap filler, schema integrity validator, vision pass, and Google Slides agent — for intelligent textbook processing at scale.
- Built a Seminar PPT Automation Co-pilot (LangGraph stateful graphs + Qdrant RAG + MCP) that monitors student Google Slides edits in real time, performs RAG-based validation, and applies corrections with human-in-the-loop approval gates.
- Engineered a high-accuracy PDF OCR & Enrichment Pipeline using Mistral OCR and GPT-4o-mini; integrated a vision fallback agent (GPT-4o Vision) for low-quality pages and an adaptive schema extractor handling any subject without hardcoded rules.

- Developed a Hybrid RAG system combining Qdrant vector search (Voyage AI embeddings) with SearXNG web search, powering AI Tutor, multi-user Debate Engine, Seminar Engine, and English Teacher with real-time semantic retrieval.
- Implemented per-student Google OAuth via Scalekit SDK and batchUpdate Google Slides API calls enabling fully autonomous AI-driven slide editing on each student's presentation.
- Built complete Avatar Engine, multi-user Debate & Seminar engines, English Teacher, Quiz, and Question Bank systems with 40+ FastAPI REST endpoints.
- Implemented adaptive learning: topic-aware homework auto-assignment (triggered after 10 tutor queries on same topic), AI-based quiz grading, and unified student performance timeline.
- Integrated end-to-end observability with Langfuse (token usage, latency, quality monitoring); deployed on AWS S3 and Qdrant Cloud.

MAX — AI Powered Chatbot *[Multi-Domain]*

- Led development of MAX, an AI-powered chatbot offering five capabilities: resume generator, disease prediction model, general chatbot, Gita AI, and personal assistant.
- Integrated advanced AI functionalities across multiple domains, delivering intelligent, context-aware responses to diverse user queries.

Automated Media Player *[Computer Vision]*

- Built a gesture-controlled media player using Python and OpenCV that controls YouTube, MX Player, and VLC via real-time hand landmark detection.
- Mapped gestures to media commands (play/pause, volume, seeking, mute) via keyboard shortcut injection for seamless hands-free control.

EDUCATION

Bachelor of Engineering — Computer Science *GPA: 7.6*

Kingston Engineering College, Vellore, Tamil Nadu *Aug 2020 – Jul 2024*

CERTIFICATIONS

- AI & ML, Data Science Bootcamp — Supervised / Unsupervised / Reinforcement Learning, Deep Learning (CNNs, RNNs, Transformers), NLP, LLMs, Generative AI. Frameworks: TensorFlow, PyTorch, Keras.
- Data Visualization — Empowering Business with Effective Insights — Transforming complex datasets into actionable dashboards and visuals to support data-driven decision-making and strategic planning.